



BCA FINAL YEAR PROJECT — 2023–2026

Student Training Management System

A modern web-based application for managing student records, training programs, company placements, and academic results.

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Under the guidance of **Mr. S.K Ojha**, Assistant Professor

Sityog Institute of Technology, Aurangabad (Bihar) — Session 2023–2026

Department of Computer Application — Bachelor of Computer Application

#	Name	Registration No.	Role
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Project Guide

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What is this system?

The Student Training Management System is a web-based application designed to efficiently maintain and manage student records in colleges, universities, and training institutes.

What problem does it solve?

Educational institutions traditionally maintain records manually — which is time-consuming, error-prone, and difficult to update. This system provides a centralized digital platform to overcome those challenges.

What does it manage?

- Personal and academic details of students
- Company and placement information
- Training and internship records
- Academic results and grades
- Comprehensive reports for all modules

Key Benefits

- Reduces paperwork and manual effort
- Centralised and secure data storage
- Quick search and retrieval
- Accessible from any browser

- **Student Record Management** — Maintain and manage student personal details, academic records, training details, and project information in a centralised database.
- **Training & Company Management** — Store and manage information related to training programs, companies, internships, and placement activities.
- **Efficient Search & Retrieval** — Provide facilities for quick searching, updating, and retrieval of student and training records whenever required.
- **Project & Result Tracking** — Maintain records of student projects, training outcomes, and academic results for effective monitoring.
- **Report Generation** — Generate various reports including student reports, training records, result summaries, and individual student profiles.
- **Secure Access Control** — Implement authentication mechanisms so only authorised administrators can access and modify sensitive data.

Layer	Technology	Purpose
Framework	Next.js 16	Full-stack React framework
Language	TypeScript	Type-safe development
Styling	Tailwind CSS	Utility-first CSS framework
UI Components	shadcn/ui	Modern component library
Database	PostgreSQL	Relational data storage
Cloud DB	Neon DB	Serverless PostgreSQL
ORM	Prisma 7	Database access layer
Auth	NextAuth v5	Session management

Why Next.js over VB.NET?

- Works on all platforms (macOS, Windows, Linux)
- Web-based — accessible from any browser
- Modern, industry-relevant stack
- Aligns with Recommendation #1 in our synopsis (web-based upgrade)

Why Neon DB over SQL Server?

- Free serverless PostgreSQL — no setup needed
- Scales automatically
- Works seamlessly with Prisma ORM
- PostgreSQL is industry standard

 Authentication

- Admin login/logout
- JWT session management
- Route protection
- Credential validation

 Dashboard

- Summary statistics
- Total students/companies
- Training & result counts
- Recent student records

 Student Module

- Search by name / roll no.
- Add / edit / delete records
- View individual profile
- Training & result history

 Company Module

- Search by company name
- Add / edit / delete records
- Contact & industry info
- Placement tracking

 Training Module

- Search by student name
- Add / edit / delete records
- Status: Ongoing / Completed
- Project description

 Results Module

- Search by student name
- Add / edit / delete records
- Marks & grade tracking
- Semester-wise records

Reports Module — View all students, companies, training records, and results in a single consolidated page with summary statistics.



Student Management System

Sityog Institute of Technology, Aurangabad

Sign in

Enter your admin credentials to continue

Username

Password

Sign in

Default credentials: **admin** / **admin123**

 **Student MS**
SIT Aurangabad

Dashboard

Students

Companies

Training

Results

Reports

N 1 Issue X

Dashboard

Welcome to Student Training Management System

Total Students

5

Companies

3

Training Records

3

Results

5

Recent Students

View all

Roll No.	Name	Course	Year	Email
BCA012	Himanshu Kumar Singh	BCA	Year 3	krhimanshuet766@gmail.com
BCA004	Om Prakash Kumar	BCA	Year 3	om@example.com
BCA003	Mohit Raj	BCA	Year 3	mohit@example.com
BCA002	Chirag Singh	BCA	Year 3	chirag@example.com
BCA001	Abhishek Kumar	BCA	Year 3	abhishek@example.com

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Dashboard

Students

Companies

Training

Results

Reports

N

 1 Issue

×

Students

Manage all student records

+ Add Student

Roll No.	Name	Email	Phone	Course	Year	Actions
BCA012	Himanshu Kumar Singh	krhimanshuet766@gmail.com	8368346653	BCA	Year 3	  
BCA004	Om Prakash Kumar	om@example.com	9876543213	BCA	Year 3	  
BCA003	Mohit Raj	mohit@example.com	9876543212	BCA	Year 3	  
BCA002	Chirag Singh	chirag@example.com	9876543211	BCA	Year 3	  
BCA001	Abhishek Kumar	abhishek@example.com	9876543210	BCA	Year 3	  

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 Dashboard

 Students

 **Companies**

 Training

 Results

 Reports

 Logout

Companies

Manage company and placement records

Company	Industry	Location	Contact Person	Email	Phone	Actions
Tech mahindra	IT	Hyd	Himanshu Kumar Singh	krhimanshuet766@gmail.com	8368346653	 
Infosys Ltd.	IT Services	Bangalore, Karnataka	Priya Mehta	hr@infosys.com	9002345678	 
TechCorp Pvt. Ltd.	IT	Patna, Bihar	Rajan Sharma	hr@techcorp.com	9001234567	 

+ Add Company

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 Dashboard

 Students

 Companies

 **Training**

 Results

 Reports

 Logout

Training

Manage internship and training records

+ Add Training

Search by student name or roll no...

Student	Company	Project	Duration	Status	Actions
Himanshu Kumar Singh BCA012	Infosys Ltd.	Python	1/16/2026 – 7/16/2026	Completed	 
Abhishek Kumar BCA001	TechCorp Pvt. Ltd.	Inventory Management System	6/1/2025 – 8/31/2025	Completed	 
Chirag Singh BCA002	Infosys Ltd.	HR Portal	7/1/2025 – 9/30/2025	Ongoing	 

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Dashboard

Students

Companies

Training

Results

Reports

Logout

Results

Manage academic results and grades

+ Add Result

Q Search by student name...

Student	Subject	Marks	Grade	Semester	Year	Actions
Abhishek Kumar BCA001	Web Technology	90/100	A+	Sem 5	2025	 
Chirag Singh BCA002	DBMS	78/100	B+	Sem 5	2025	 
Mohit Raj BCA003	DBMS	82/100	A	Sem 5	2025	 
Om Prakash Kumar BCA004	Web Technology	88/100	A	Sem 5	2025	 
Abhishek Kumar BCA001	DBMS	85/100	B+	Sem 5	2025	 

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Dashboard

Students

Companies

Training

Results

Reports

Reports

Overview of all records

Total Students

5

Total Companies

3

Trainings Completed

2

Trainings Ongoing

1

All Students

5 records

Roll No.	Name	Course	Year	Email	Phone
BCA001	Abhishek Kumar	BCA	Year 3	abhishek@example.com	9876543210
BCA002	Chirag Singh	BCA	Year 3	chirag@example.com	9876543211
BCA012	Himanshu Kumar Singh	BCA	Year 3	krhimanshuet766@gmail.com	8368346653
BCA003	Mohit Raj	BCA	Year 3	mohit@example.com	9876543212
BCA004	Om Prakash Kumar	BCA	Year 3	om@example.com	9876543213

All Companies

3 records

Company	Industry	Location	Contact	Trainings
Infosys Ltd.	IT Services	Bangalore, Karnataka	Priya Mehta	2
Tech mahindra	IT	Hyd	Himanshu Kumar Singh	0
TechCorp Pvt. Ltd.	IT	Patna, Bihar	Rajan Sharma	1

Admin Table

- id, username, password, createdAt

Student Table

- id, rollNo (unique), name, email
- phone, address, course, year
- createdAt, updatedAt

Company Table

- id, name (unique), industry
- location, contactPerson
- email, phone, createdAt

Training Table

- id, studentId (FK), companyId (FK)
- projectName, startDate, endDate
- status, description

Result Table

- id, studentId (FK)
- subject, marks, maxMarks, grade
- semester, year

Relationships

- Student → Training (one-to-many)
- Company → Training (one-to-many)
- Student → Result (one-to-many)
- Cascade delete on Student/Company

Key Features Implemented

- Secure admin authentication with JWT
- Full CRUD for all 4 modules
- Real-time search and filtering
- Skeleton loading states for better UX
- Side drawer for add/edit forms
- Individual student profile with history
- Consolidated reports page
- Cloud PostgreSQL (Neon DB)
- Responsive modern UI with shadcn/ui

Future Scope (Recommendations)

- Role-based access (Admin, Faculty, Student)
- Email/SMS notification system
- Mobile application support
- Online result viewing for students
- Placement & internship tracking dashboard
- PDF export for individual reports
- Attendance management module
- Multi-campus support

Thank You

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